EM5033 : TEXT ANALYTICS FOR BUSINESS

Final Project

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About Ixigo

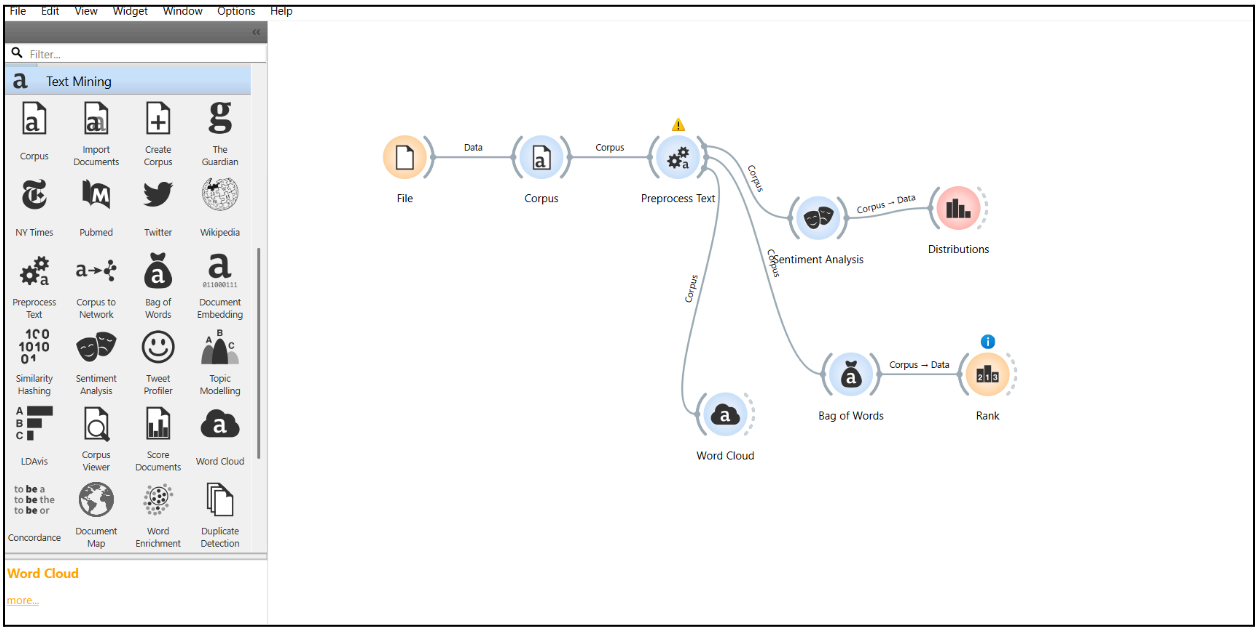
Ixigo is an Indian travel technology company founded in 2007 by Aloke Bajpai and Rajnish Kumar, headquartered in Haryana. Operating under its parent entity Le Travenues Technology Limited, Ixigo functions as an AI-powered online travel aggregator that enables users to search, compare, and book flights, trains, buses, and hotels across India. The platform is particularly well-known for its deep integration with Indian Railways data, offering real-time PNR status tracking, train running status, and seat availability , features that have made it especially popular among budget and tier-2/tier-3 city travellers. With over 10 crore (100 million) downloads on Google Play and a strong presence in non-metro India, Ixigo has carved a niche as an accessibility-first travel platform that caters to the price-sensitive, first-time digital traveller. In June 2024, Ixigo successfully listed on the National Stock Exchange (NSE) and Bombay Stock Exchange (BSE), marking a significant milestone in its growth trajectory. The company competes in a crowded OTA (Online Travel Agency) market alongside players such as MakeMyTrip, EaseMyTrip, Yatra, and Cleartrip, making user trust and platform reliability critical differentiators for sustaining long-term market share and revenue growth.

Aim of the Project

The aim of this project is to conduct a comprehensive sentiment analysis of user reviews posted on Google Play for the Ixigo travel application, with the objective of deriving actionable business insights that can inform strategies for improving customer satisfaction and driving revenue growth. Using Orange, an open-source data mining and text analytics tool, the project preprocesses and analyses a dataset of 9,609 user-generated reviews to identify dominant sentiment patterns, recurring pain points, and thematic complaint clusters. By applying natural language processing techniques, including tokenization, stop-word removal, and sentiment classification , it seeks to move beyond surface-level star ratings and uncover the specific drivers of negative user sentiment. The findings are then contextualized within a business framework to evaluate how unresolved issues such as refund failures, pricing transparency, and customer support gaps translate into measurable impacts on user retention, acquisition costs, and lifetime value.

Data Preprocessing in Orange

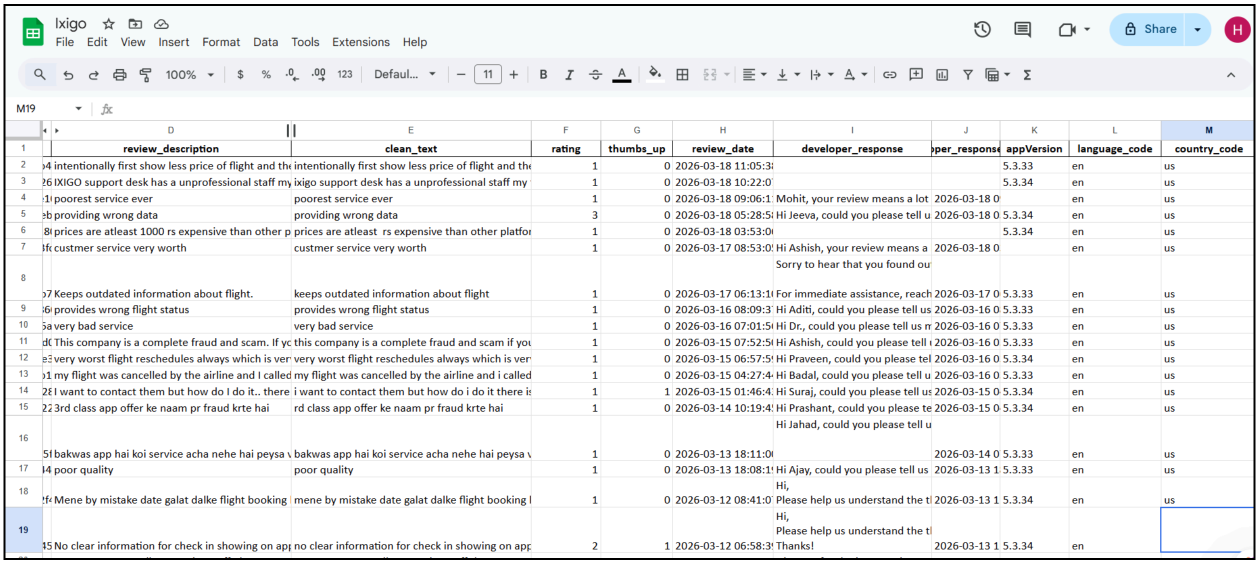
Before any sentiment analysis could be performed, the raw review data extracted from Google Play required a series of structured preprocessing steps carried out using Orange's **Text Mining add-on**. These steps were essential to convert unstructured, noisy user-generated text into a clean and analyzable format suitable for machine-driven sentiment classification and keyword analysis.



Orange data flow

Step 1: Data Import using the CSV File Import Widget

The first step involved loading the extracted Google Play review dataset into Orange using the CSV File Import widget. The dataset, containing 9,609 reviews across 15 variables including review description, star rating, thumbs-up count, and review date, was imported into the Orange workflow. The relevant column for analysis, **review_description** , was identified as the primary text field, while **rating** served as the target variable for supervised and comparative analysis.



	review_description	clean_text	rating	thumbs_up	review_date	developer_response	per_response	appVersion	language_code	country_code
2	intentionally first show less price of flight and the	intentionally first show less price of flight and the	1	0	2026-03-18 11:05:3			5.3.33	en	us
3	IXIGO support desk has a unprofessional staff my	ixigo support desk has a unprofessional staff my	1	0	2026-03-18 10:22:0			5.3.34	en	us
4	poorest service ever	poorest service ever	1	0	2026-03-18 09:06:1	Mohit, your review means a lot	2026-03-18 0		en	us
5	providing wrong data	providing wrong data	3	0	2026-03-18 05:28:5	Hi Jeeva, could you please tell u	2026-03-18 0	5.3.34	en	us
6	prices are atleast 1000 rs expensive than other p	prices are atleast rs expensive than other platfor	1	0	2026-03-18 03:53:0			5.3.34	en	us
7	custmer service very worth	custmer service very worth	1	0	2026-03-17 08:53:0	Hi Ashish, your review means a	2026-03-18 0		en	us
8	Keeps outdated information about flight.	keeps outdated information about flight	1	0	2026-03-17 06:13:1	Sorry to hear that you found ou For immediate assistance, reach	2026-03-17 0	5.3.33	en	us
9	provides wrong flight status	provides wrong flight status	1	0	2026-03-16 08:09:3	Hi Aditi, could you please tell us	2026-03-16 0	5.3.33	en	us
10	very bad service	very bad service	1	0	2026-03-16 07:01:5	Hi Dr., could you please tell us m	2026-03-16 0	5.3.33	en	us
11	This company is a complete fraud and scam. If yo	this company is a complete fraud and scam if you	1	0	2026-03-15 07:52:5	Hi Ashish, could you please tell	2026-03-16 0	5.3.33	en	us
12	very worst flight reschedules always which is ver	very worst flight reschedules always which is ver	1	0	2026-03-15 06:57:5	Hi Praveen, could you please tel	2026-03-16 0	5.3.34	en	us
13	my flight was cancelled by the airline and I called	my flight was cancelled by the airline and i called	1	0	2026-03-15 04:27:4	Hi Badal, could you please tell u	2026-03-16 0	5.3.33	en	us
14	I want to contact them but how do I do it.. there	i want to contact them but how do i do it there is	1	1	2026-03-15 01:46:4	Hi Suraj, could you please tell us	2026-03-15 0	5.3.34	en	us
15	3rd class app offer ke naam pr fraud krte hai	rd class app offer ke naam pr fraud krte hai	1	0	2026-03-14 10:19:4	Hi Prashant, could you please te	2026-03-15 0	5.3.34	en	us
16	bakwas app hai koi service acha nehe hai peysa v	bakwas app hai koi service acha nehe hai peysa v	1	0	2026-03-13 18:11:0	Hi Jahad, could you please tell u	2026-03-14 0	5.3.33	en	us
17	poor quality	poor quality	1	0	2026-03-13 18:08:1	Hi Ajay, could you please tell us	2026-03-13 1	5.3.33	en	us
18	Mene by mistake date galat dalke flight booking	mene by mistake date galat dalke flight booking	1	0	2026-03-12 08:41:0	Hi, Please help us understand the t	2026-03-13 1	5.3.34	en	us
19	No clear information for check in showing on ap	no clear information for check in showing on app	2	1	2026-03-12 06:58:3	Hi, Please help us understand the t Thanks!	2026-03-13 1	5.3.34	en	

The reviews imported to sheets after using python code that extracted it from google play

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Step 2: Defining the Text Variable using the Corpus Widget

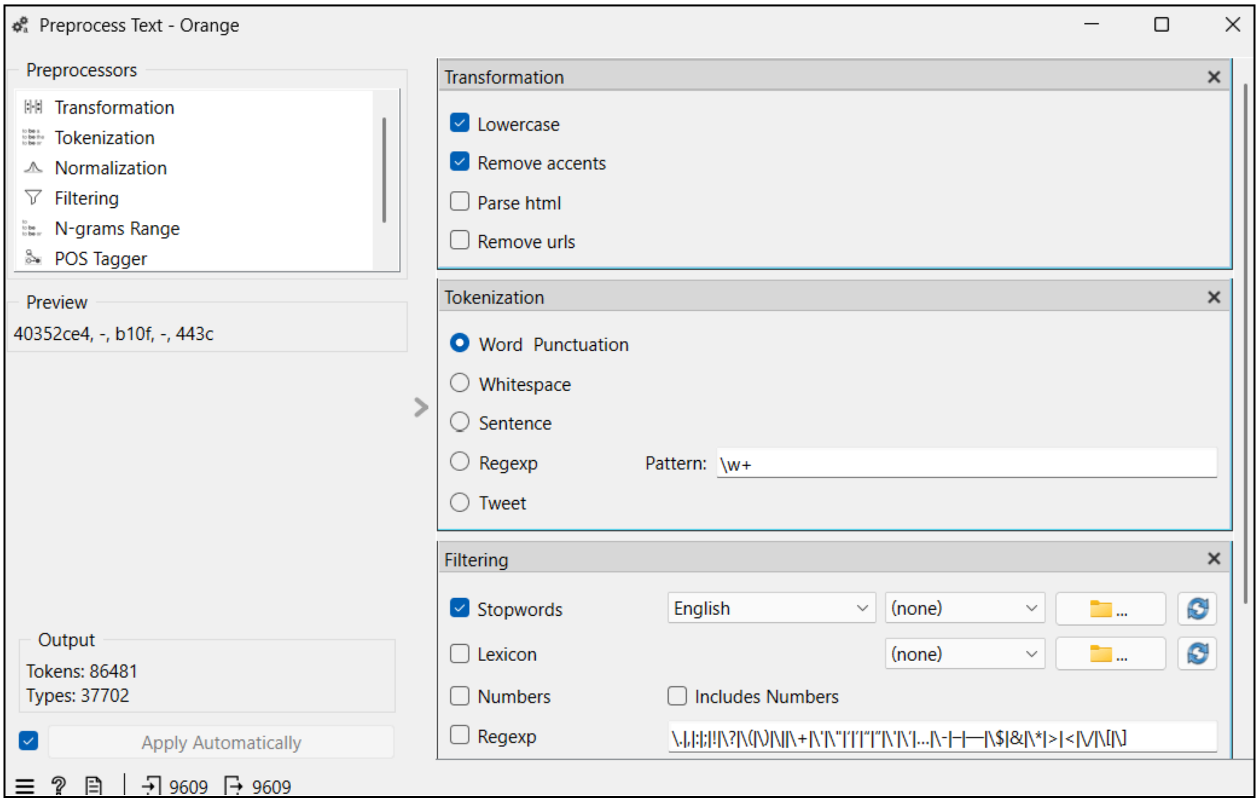
Once the dataset was loaded, the Corpus widget was used to formally define which column contained the text data to be analysed. The `review_description` column was set as the document text, and the `rating` column was retained as a metadata variable. This step converted the tabular dataset into an Orange Corpus object , the required format for all subsequent text mining operations in Orange.

Step 3: Text Preprocessing using the Preprocess Text Widget

This was the most critical and multi-layered step in the pipeline. The **Preprocess Text widget** in Orange was applied with the following sequential transformations:

- **Lowercase Transformation:** All text was converted to lowercase to ensure that the same words in upper and lower cases were treated as the same token and not counted separately.
- **Tokenization:** The continuous text of each review was split into individual tokens (words) using a word-level tokenizer.
- **Stop Word Removal:** Commonly occurring English words that carry no analytical value were removed using Orange's built-in English stop word list. This prevented high-frequency but meaningless words from dominating the analysis and allowed substantively important words like *"refund", "fraud",* and *"cancel"* to surface more prominently.

- **Punctuation and Number Removal:** Special characters, punctuation marks, and numerical values were stripped from the text to further reduce noise.
- **Normalization — Lemmatization (or Stemming):** Words were reduced to their base or root form In Orange, this is typically done through lemmatization using the built-in language model.

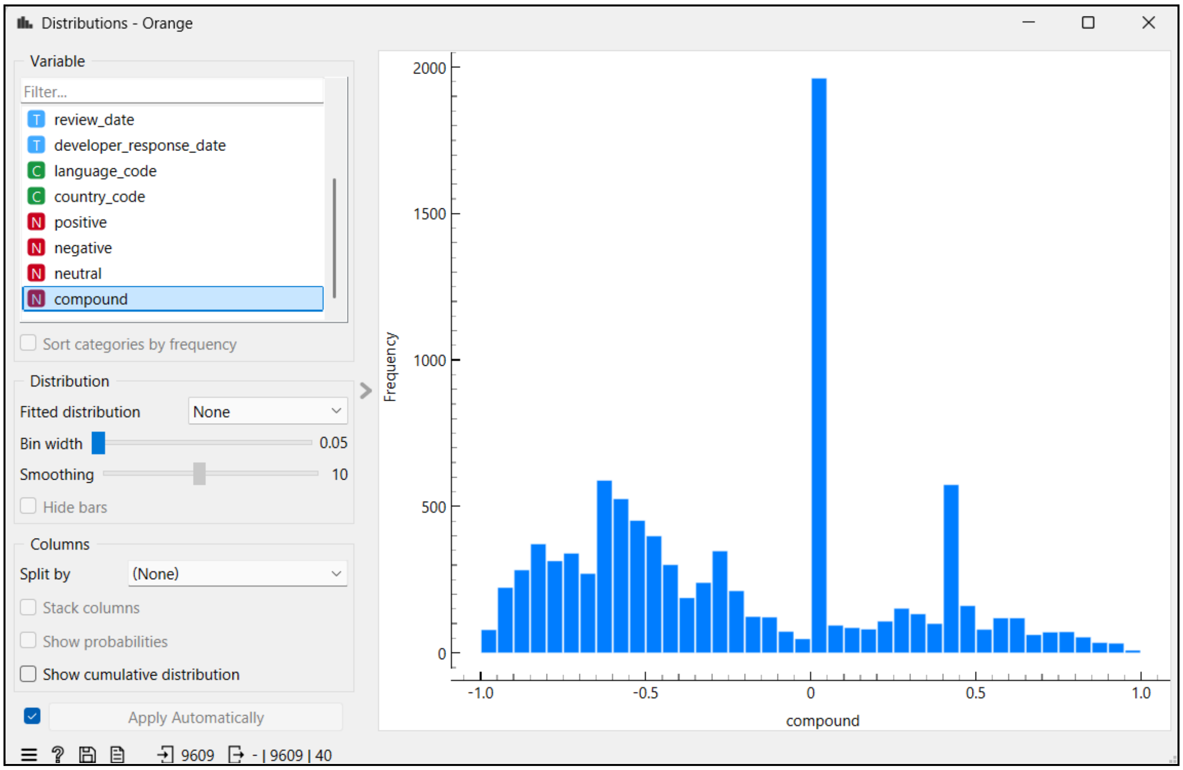


Step 5: Word Cloud / Bag of Words Representation

After preprocessing, the **Bag of Words widget** was used to convert the cleaned tokens into a numerical matrix where each review is represented as a vector of word frequencies. This representation is what makes the text mathematically processable by machine learning and sentiment models. The resulting matrix captured how often each significant word appeared across all reviews, forming the foundation for frequency-based and model-based analysis.

Step 6: Sentiment Analysis using the Sentiment Analysis Widget

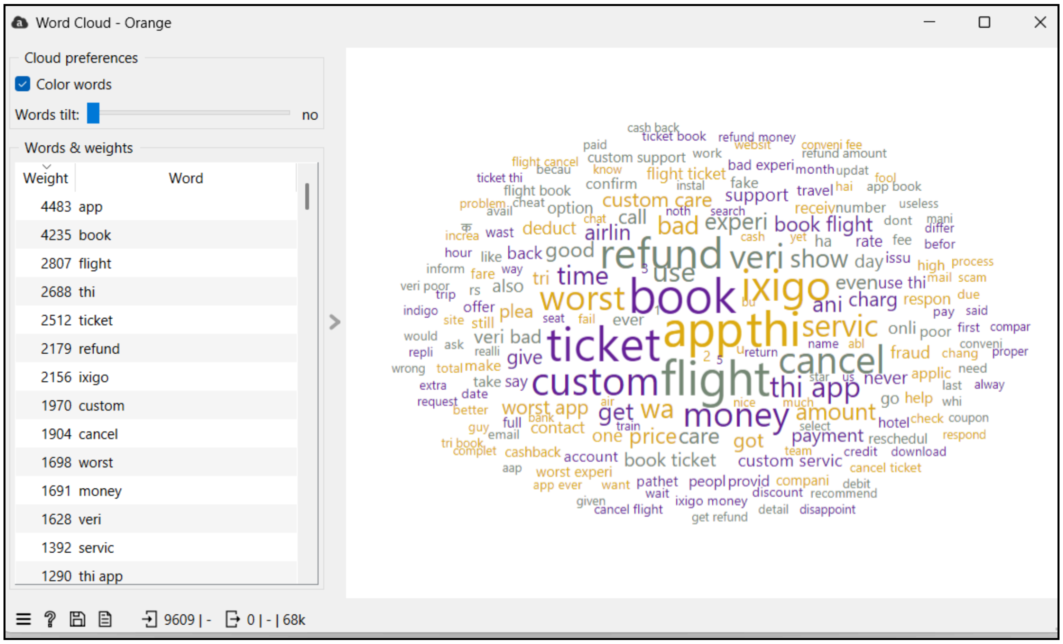
Orange's built-in **Sentiment Analysis widget** was then applied to the preprocessed corpus. This widget uses established lexicon-based sentiment scoring methods — primarily **VADER (Valence Aware Dictionary and sentiment Reasoner)**, which is well-suited for short, informal, user-generated text like app reviews. VADER assigns each review a **compound sentiment score** ranging from -1 (most negative) to +1 (most positive), along with individual positive, negative, and neutral sub-scores. These scores were then used to classify reviews into sentiment categories and correlate them with star ratings for validation.



Distribution of sentiment

Step 7: Visualization using Word Cloud and Distributions Widget

Finally, Orange's Word Cloud widget was used to visually represent the most frequently occurring terms in the preprocessed corpus — with word size proportional to frequency. Separate word clouds were generated for 1-star and higher-rated reviews to contrast the vocabulary of dissatisfied versus satisfied users. The Distributions widget was additionally used to visualise the spread of sentiment scores and star ratings across the dataset.



Word cloud

Together, these preprocessing steps formed a complete and systematic pipeline that transformed 9,609 raw, messy user reviews into a structured, clean, and analytically meaningful corpus , making the subsequent sentiment classification and business inference possible.

Dataset Overview

The dataset contains **9,609 Google Play reviews** for Ixigo spanning from **2011 to March 2026**, with 15 variables including review text, star ratings, thumbs-up counts, developer responses, and app version details.The analysis was conducted using Orange's text mining and sentiment analysis pipeline.

Overall Sentiment Distribution & What It Means

Rating	Count	% of Total
1-star	7,751	80.7%
2-star	721	7.5%
3-star	1,137	11.8%

The average rating hovers around 1.2 stars — one of the lowest possible scores. This is a major dissatisfaction signal. In OTA (Online Travel Agency) markets like India where MakeMyTrip, EaseMyTrip, and Yatra are direct competitors, a persistently negative app store reputation directly impacts organic downloads, conversion rates, and customer lifetime value.It's worth noting that the dataset skews heavily negative, partly because satisfied users are less likely to leave reviews,so the actual ratio of happy to unhappy users may not be as extreme as the numbers suggest.Since App store ratings are a major discovery and trust signal. Studies consistently show that apps below 3.5 stars see significantly lower install rates.

Dominant Pain Point Themes (Sentiment Drivers)

Refund & Cancellation Failures

Words like refund, cancel, cancellation, money back dominate. Reviews describe scenarios where:

- Refunds promised within "2–3 business days" never arrive for weeks or months
- Cancellation portals show "processing" indefinitely with no resolution
- Users are forced to dispute through their bank or credit card company

Delayed refunds destroy repeat purchase intent. In travel, where a single customer can generate 4–8 bookings per year, losing them to a competitor after one bad refund experience has a negative impact. A 10% improvement in refund resolution speed could materially reduce churn in this segment.

Price Deception & Hidden Charges

Keywords: price, fare, hidden charge, expensive, cheat, fraud, mislead

Reviewers consistently describe a pattern:

1. Ixigo displays an attractively low fare to attract clicks
2. At the final checkout step, the price increases ("fare alert")
3. Additional charges — fuel charges, convenience fees, taxes — inflate the final amount far beyond what was shown

Price transparency is not just an ethical issue, it prevents people from trusting the app. Bait-and-switch pricing creates cart abandonment, negative word of mouth, and negative review. Competitors who display all-inclusive pricing upfront win the trust of the customer. Ixigo risks regulatory scrutiny as well, given India's growing consumer protection framework around digital commerce.

Customer Support Inaccessibility

Keywords: customer service, support, helpline, contact, response
A recurring complaint is that Ixigo makes it structurally difficult to reach a human agent:

- No visible phone number or direct email in the app
- Automated responses directing users to FAQ articles that don't resolve their issue
- Chat support described as slow, scripted, or non-functional

I noticed that despite 99.9% of reviews receiving a developer response, these responses are generic ("Please contact us at...") and do not resolve the underlying issue, as seen in the follow-up complaints in the same threads.

A high developer response rate without issue resolution creates the appearance of engagement while delivering no customer value.In contrast, platforms like MakeMyTrip have invested in 24/7 live chat, which directly reduces escalations and negative reviews.

Payment & Transaction Failures

Keywords: payment, transaction, deducted, bank, amount

Users report money being debited from their accounts with:

- No booking confirmation generated
- No automatic reversal from the bank
- Support refusing to acknowledge the transaction

Payment failure without automatic reversal is a regulatory and reputational risk. Under RBI guidelines, failed transactions must be reversed within a defined timeframe. Non-compliance creates legal exposure. More practically, a user who loses money, even temporarily will never return, and will actively warn others. This alone accounts for a measurable percentage of uninstalls.

App & Technical Issues

Keywords: *server error, crash, loading, slow, bug*

While only a few reviews explicitly mention technical issues, server errors are embedded in refund failures, payment issues, and booking problems ,making this an underreported but amplifying factor across all other pain points.

Technical instability during high-demand periods (festival seasons, long weekends) is when booking volume and therefore revenue opportunity is highest. Server failures during peak traffic directly translate into lost profits.

Recommendations

1. Fix the Refund Process Before It Fixes Ixigo's Reputation The loudest complaint across thousands of reviews is money not coming back. Ixigo should build a transparent, automated refund tracking system where users can see exactly where their refund stands, processed, in transit, or credited without having to chase support. When people can see progress, they stop panicking, stop writing angry reviews, and are far more likely to book again. A customer who got their refund smoothly is a repeat customer.

2. Stop Showing One Price and Charging Another Ixigo's current practice of displaying an attractive base fare and then adding fuel surcharges, convenience fees, and taxes at the final step is the root cause of the "fraud" label users repeatedly apply to the app. Indigo needs to show the total, all-inclusive price from the very first search result. Competitors who do this are winning the trust of customers who have already been burned by Ixigo. Honest pricing doesn't just improve reviews; it improves the conversion rate because users stop abandoning checkout out of frustration.

3. Give People an Actual Human to Talk To Ixigo's current support experience , FAQ articles, templated bot responses, and no visible phone number or email, is not support, it is avoidance. When someone's money is stuck or their flight has been cancelled, they do not want to be redirected to a help article. Adding a real in-app live chat staffed by trained agents, especially during high-volume travel periods, would directly reduce the volume of one-star reviews that stem purely from the inability to reach anyone.

4. When a Payment Fails, Own It Immediately Losing money to a failed transaction with no confirmation and no automatic reversal is one of the most distressing things that can happen to a user on any platform. Ixigo needs to implement same-day or next-day automatic reversals for failed transactions, paired with a proactive SMS or in-app notification confirming the reversal is on its way. It’s not just about user experience but about regulatory compliance and basic financial trust. A user who loses even ₹500 to a failed booking and hears nothing will never return, and will make sure their network knows about it.

5. The App Cannot Break When It Matters Most Server errors and app crashes are disproportionately reported during peak booking windows , holiday seasons, long weekends, and last-minute travel surges which are precisely the moments when Ixigo stands to earn the most. Investing in infrastructure that can handle traffic spikes, through optimization and rigorous load testing ahead of known peak periods, is not a technical nicety but a direct revenue protection measure.

Conclusion

The sentiment analysis of Ixigo's Google Play reviews presents a clear picture: the problems users face are not random or occasional, they keep coming up in the same areas, which suggests they haven't been fixed. The five core pain points - **refunds, pricing transparency, customer support, payment failures, and technical instability** are interconnected and dependent. Addressing a few of these properly could shift the rating above 3.0, which represents a positive effect for the app and can also gain user trust will create a better word of mouth.

For a company competing in India's rapidly growing travel market, user sentiment is not just a metric, it is a direct driver of Customer Acquisition Cost, Lifetime Value, and market share. The data strongly suggests that Ixigo's current revenue growth is being achieved despite its user experience, and that resolving these issues represents the highest investment the company can make.

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